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Panel mount electrical connector

Abstract

An electrical connector includes a connector housing having a base, a flange, and a plug. The connector housing has a contact channel holding a contact. The electrical connector includes a latch member integral with the connector housing and located within the flange. The latch member includes a deflectable latch arm and a latching finger extending from the latch arm. The plug is configured to be inserted into a panel opening of the panel in a loading direction. The connector housing is movable laterally relative to the panel in a latching direction to a latched position. The plug is movable in the panel opening in the latching direction. The latching finger engages an inner edge of the panel opening when the housing is in the latched position to hold the connector housing in the latched position.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims benefit to U.S. Provisional Application No. 63/253,776, filed 8 Oct. 2021, titled "PANEL MOUNT ELECTRICAL CONNECTOR", the subject matter of which is herein incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION
(1) The subject matter herein relates generally to electrical connectors.
(2) In general, electrical connectors are used to electrically connect components. The electrical connectors may transmit data signals and/or power between the electrical components. Typically, the electrical connector includes one or more contacts held in a housing. The electrical connector is mated to a mating component, such as another electrical connector or a busbar.
(3) In at least some electronic systems, the electrical connectors are mounted to a connector panel for mating with a complementary mating connector. The connectors are mounted to the connector panel using a variety of methods such as brackets, clamps, threaded bolts, or other fasteners. The additional securing components add to the cost and complexity of assembly. Additionally, with the continuously increasing demand for resources in today's systems, connector space on the connector panels is in short supply. In many instances, due to space limitations, system operators limit the amount of connector space available for each application. In addition to the size of the connectors, features that may be provided for particular mounting arrangements may also contribute to space

shortages on the connector panel by increasing the space required between connectors.
(4) A need remains for an electrical connector having a small footprint to facilitate saving space on the connector panels. It would be further desirable to provide an electrical connector that is mountable to the panel without the need for tools or mounting hardware.
BRIEF DESCRIPTION OF THE INVENTION
(5) In one embodiment, an electrical connector is provided for mounting to a panel. The electrical connector includes a connector housing having a base, a flange, and a plug. The connector housing has a contact channel holding a contact. The electrical connector includes a latch member integral with the connector housing and located within the flange. The latch member includes a deflectable latch arm and a latching finger extending from the latch arm. The plug is configured to be inserted into a panel opening of the panel in a loading direction. The connector housing is movable laterally relative to the panel in a latching direction to a latched position. The plug is movable in the panel opening in the latching direction. The latching finger engages an inner edge of the panel opening when the housing is in the latched position to hold the connector housing in the latched position.
(6) In another embodiment, an electrical connector system includes a panel and an electrical connector mounted to the panel. The panel has an inner edge defining a panel opening. The panel opening includes a main pocket and a latching pocket extending from the main pocket. The electrical connector includes a connector housing that has a base at a rear of the connector housing, a flange extending from the base, and a plug extending forward of the base and the flange at a front of the connector housing for mating with a mating electrical connector. The connector housing has a contact channel through the base and the plug. The electrical connector includes a contact received in the contact channel having a mating end and a cable end configured to be terminated to a cable. The electrical connector includes a latch member integral with the connector housing and located within the flange. The latch member includes a latch arm being deflectable relative to the connector housing. The latch member includes a latching finger extending from the latch arm. An electrical connector system includes the plug of the connector housing that is inserted into the main pocket of the panel opening in a loading direction. The connector housing is movable laterally relative to the panel in a latching direction to a latched position. The latching direction differs from the loading direction. The plug is movable in the main pocket in the latching direction. The latching finger is received in the latch pocket in the latched position and engaging the inner edge of the panel opening in the latched position to hold the connector housing in the latched position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** illustrates an electrical connector system in accordance with an exemplary embodiment.
(2) FIG. **2** is a front perspective view of the electrical connector in accordance with an exemplary embodiment.
(3) FIG. **3** is a front perspective view of a portion of the electrical connector showing the connector housing in accordance with an exemplary embodiment
(4) FIG. **4** is a rear perspective view of a portion of the electrical connector showing the connector housing with an exemplary embodiment.
(5) FIG. **5** is a sectional view of a portion of the electrical connector in accordance with an exemplary embodiment.
(6) FIG. **6** is a top sectional view of a portion of the electrical connector in accordance with an exemplary embodiment.
(7) FIG. **7** illustrates the electrical connector system showing the electrical connector poised for mounting to the panel with an exemplary embodiment.
(8) FIG. **8** illustrates the electrical connector system showing the electrical connector partially

mated to the panel with an exemplary embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(9) FIG. **1** illustrates an electrical connector system **100** in accordance with an exemplary embodiment. The electrical connector system **100** includes a panel mounted electrical connector **200** mounted to a panel **102**. The electrical connector **200** is configured to be electrically connected to a mating electrical connector **104**. In the illustrated embodiment, the mating electrical connector **104** is a power connector configured to supply power to the electrical connector **200**. For example, the mating electrical connector **104** includes a busbar **106** supplying power to the electrical connector **200**. Other types of mating electrical connectors **104** may be used in alternative embodiments, such as a data communication connector configured to transmit electrical signals between the mating electrical connector **104** and the electrical connector **200**. In various embodiments, the mating electrical connector **104** may transmit power and data to the electrical connector **200**.

(10) The panel **102** may be a chassis, a frame, a housing, or other component of the electrical connector system **100**. In an exemplary embodiment, the panel **102** is planar having a front surface **110** and the rear surface **112**. In various embodiments, the panel **102** is electrically conductive and may be electrically grounded. For example, the panel **102** may be a piece of sheet metal. The electrical connector **200** may be electrically communed with the panel **102**.

(11) The panel **102** includes a panel opening **120** therethrough. The electrical connector **200** is mounted to the panel **102** at the panel opening **120**. For example, a portion of the electrical connector **200** may pass through the panel opening **120** for mating with the mating electrical connector **104**. In an exemplary embodiment, a portion of the electrical connector **200** is coupled to the rear surface **112** and a portion of the electrical connector **200** is coupled to the front surface **110**. The panel **102** is captured between features of the electrical connector **200** such that the electrical connector **200** is fixed in position at the panel **102** for mating with the mating electrical connector **104**.

(12) In an exemplary embodiment, the electrical connector **200** may be latchably coupled to the panel **102**. For example, the electrical connector **200** includes one or more latching features that are latchably coupled to the panel **102**. In an exemplary embodiment, the electrical connector **200** is mounted to the panel **102** without the use of tools or mounting hardware. In an exemplary embodiment, the electrical connector **200** is a slide-to-latch electrical connector configured to be loaded through the panel opening **120** to a nominal position and then slid in a latching direction to a latched position. Integral features of the electrical connector **200** engage the panel **102** to secure the electrical connector **200** to the panel **102**.

(13) The panel opening **120** is defined by an inner edge **122** surrounding the panel opening **120**. In the illustrated embodiment, a single panel opening **120** is provided. However, the panel **102** may include multiple panel openings **120** in alternative embodiments. In an exemplary embodiment, the inner edge **122** is polygonal, having a plurality of straight-line segments **124** meeting at corners **126**. However, the panel opening **120** may have other shapes in alternative embodiments, including curved segments.

(14) The panel opening **120** includes a main pocket **130** and one or more side pockets **132** extending from one or more sides of the main pocket **130**. Portions of the electrical connector **200** are received in the main pocket **130** and the side pockets **132** during loading and coupling of the electrical connector **200** to the panel **102**. The side pockets **132** are cutouts or notches in the panel **102** that open to the main pocket **130**. The main pocket **130** is larger than any of the side pockets **132**. Optionally, the main pocket **130** may occupy a majority of the panel opening **120**. In an exemplary embodiment, the side pockets **132** include one or more latch pockets **134** and one or more tab pockets **136**. In the illustrated embodiment, the panel opening **120** includes a first latch pocket **134** at a first side of the main pocket **130** and a second latch pocket **134** at a second side of the main pocket **130**. The panel opening **120** may include greater or fewer latch pockets **134** in

alternative embodiments. The latch pockets **134** receive latches of the electrical connector **200**. In the illustrated embodiment, the panel opening **120** includes an upper tab pocket **136** at the top side of the main pocket **130** and a lower tab pocket **136** at a bottom side of the main pocket **130**. The panel opening **120** may include greater or fewer tab pockets **136** in alternative embodiments. The tab pockets **136** received tabs of the electrical connector **200**.

(15) FIG. **2** is a front perspective view of the electrical connector **200** in accordance with an exemplary embodiment. The electrical connector **200** includes a connector housing **202** holding one or more contacts **204**. The contacts **204** are configured to be electrically connected to the mating electrical connector **104** (shown in FIG. **1**). For example, the contacts **204** may be electrically connected to the busbar **106**. In an exemplary embodiment, the contacts **204** are provided at ends of cables **206** extending from the connector housing **202**. The connector housing **202** includes features used to secure the electrical connector **200** to the panel **102** (shown in FIG. **1**).

(16) The connector housing **202** includes a front **210** and a rear **212**. In the illustrated embodiment, the front **210** defines a mating end **214** configured to be mated with the mating electrical connector **104**. Optionally, the cables **206** may extend from the rear **212**. The connector housing **202** includes a top **216** and a bottom **218**. The connector housing **202** includes a first side **220** and a second side **222**.

(17) In an exemplary embodiment, the connector housing **202** includes a base **230** at the rear **212** and a plug **232** at the front **210**. The connector housing **202** includes a flange **234** extending from the base **230**. In various embodiments, the flange **234** may extend from the base **230** in all directions. For example, the flange **234** may include an upper flange portion **236** at the top **216**, a lower flange portion **238** at the bottom **218**, a left flange portion **240** at the left side **220**, and a right flange portion **242** at the right side **222**. The flange **234** is used for mounting the electrical connector **200** to the panel **102**. For example, the flange **234** may face the rear surface **112** of the panel **102**. The base **230** is located rearward of the flange **234**, and is thus configured to be located behind the panel **102**. The plug **232** extends forward of the flange **234**, and thus is configured to be located forward of the panel **102**. For example, the plug **232** is configured to extend through the panel opening **120** (shown in FIG. **1**) for mating with the mating electrical connector **104**.

(18) In an exemplary embodiment, the cable housing **202** includes one or more contact channels **244** that receive corresponding contacts **204**. The contact channels **244** extend into the base **230** and into the plug **232**. The contacts **204** are configured to be terminated to the cable **206** in the base portion of the contact channels **244**. The contacts **204** are configured to be mated with the mating electrical connector **104** in the plug portion of the contact channels **244**.

(19) In an exemplary embodiment, the plug **232** includes plug walls **246** forming a slot **248**. The slot **248** is open at the front **210** to receive a portion of the mating electrical connector **104**, such as the busbar **106**. The contacts **204** are exposed within the slot **248** for mating with the mating electrical connector **104**. In the illustrated embodiment, the slot **248** extend vertically from the top **216** to the bottom **218**. Example, the slot **248** is open at the top **216** and open at the bottom **218**. The slot **248** may have other shapes in alternative embodiments. In other alternative embodiments, a plurality of the slots **248** may be provided, such as individual slots for each of the contacts **204**. For example, the slots **248** may define sockets or receptacles that receive individual contacts, such as pins, of the mating electrical connector **104**. In the illustrated embodiment, the plug walls **246** are oriented vertically and provided at the first side **220** and the second side **222** of the plug **232**. Additional plug walls **246** may be provided in alternative embodiments, such as horizontally extending plug walls.

(20) In an exemplary embodiment, the electrical connector **200** includes one or more ground elements **250** coupled to the connector housing **202**. The ground elements **250** are electrically conductive. In an exemplary embodiment, the ground elements **250** are stamped and formed from a metal sheets. The ground elements **250** are configured to be electrically connected to the panel **102**

and configured to be electrically connected to the mating electrical connector **104**. The ground elements **250** are used to electrically common the mating electrical connector **104** and the panel **102**. In the illustrated embodiment, the electrical connector **200** includes a pair of the ground elements **250**, provided at the first side **220** and the second side **222**. Each ground elements **250** includes a plug wall **252** extending along the plug **232** and a flange wall **254** extending along the flange **234**. The ground element **250** includes one or more panel tabs **256** extending from the flange wall **254** configured to engage the rear surface **112** of the panel **102**. The panel tabs **256** are deflectable and extend out of the plane of the flange wall **254** to interface with the panel **102**. The ground element **250** includes one or more mating tabs **258** extending from the plug wall **254** configured to engage the mating electrical connector **104**. The mating tabs **258** are deflectable and extend out of the plane of the plug wall **252** to interface with the mating electrical connector **104**. The ground elements **250** may be secured to the connector housing **202** using clips, brackets, fasteners, or other securing elements. The ground elements **250** may have other sizes, shapes, and/or features in alternative embodiments.

(21) In an exemplary embodiment, the connector housing **202** includes one or more front tabs **260** extending from the connector housing **202** at or near the front **210**. In various embodiments, the front tabs **260** may extend from the plug **232**, such as the plug walls **246**. The front tabs **260** are configured to engage the front surface **110** of the panel **102**. In an exemplary embodiment, the front tabs **260** are integral with the connector housing **202**. For example, the front tabs **260** may be co-molded with the connector housing **202**. In an exemplary embodiment, the connector housing **202** includes an upper front tab **260** at the top **216** and a lower front tab (not shown) at the bottom **218**. Optionally, the upper and lower front tabs **260** may be offset from each other, such as having the upper front tab **260** closer to the first side **220** and the lower front tab closer to the second side **222**. The front tabs **260** are configured to be loaded through the panel opening **120** to the front side of the panel **102**. When the electrical connector **200** is slid to the side relative to the panel **102** during mating, the front tabs **260** are configured to engage the panel **102** to secure the electrical connector **200** to the panel **102**.

(22) In an exemplary embodiment, a gap **262** is formed between the corresponding front tab **260** and the flange **234**. The panel **102** is received in the gap **262**. Optionally, the gap **262** may have a width approximately equal to the thickness of the panel **102** such that the panel **102** is tightly held between the front tab **260** and the flange **234**. Optionally, slight clearance may be provided between the panel **102** and the front tab **260** and the flange **234** to avoid binding of the connector housing **202** on the panel **102**, which allows the electrical connector **200** to slide to the latched position.

(23) In an exemplary embodiment, the electrical connector **200** includes one or more latch members used to latchably secure the electrical connector **200** to the panel **102**. For example, the electrical connector **200** may include a first latch members **300** at the left side and a second latch member **302** at the right side. The latch members **300**, **302** are integral with the connector housing **202**. For example, the latch members **300**, **302** may be manufactured from the same dielectric material as the connector housing **202**. In an exemplary embodiment, the latch members **300**, **302** are co-molded with the connector housing **202**. As such, the electrical connector **200** does not require separate mounting hardware for securing the electrical connector **200** to the panel **102**. The latch members **300**, **302** are configured to be latchably coupled to the panel **102** without the use of tools. For example, the electrical connector **200** may be a slide-to-latch electrical connector, wherein the electrical connector **200** is slid sideways along the panel **102** to the latched position. The latch members **300**, **302** latchably engage the panel **102** in the latched position. In an exemplary embodiment, the latch members **300**, **302** extend forward from the flange **234** and are configured to be received in the panel opening **120**.

(24) FIG. **3** is a front perspective view of a portion of the electrical connector **200** showing the connector housing **202** in accordance with an exemplary embodiment. FIG. **4** is a rear perspective view of a portion of the electrical connector **200** showing the connector housing **202**. FIGS. **3** and **4**

illustrate the flange **234** located between the base **230** and the plug **232**. FIGS. **3** and **4** illustrate the contact channels **244** extending through the connector housing **202**, such as through the base **230** and the plug **232**.

(25) In an exemplary embodiment, the flange **234** includes the upper flange portion **236**, the lower flange portion **238**, the left flange portion **240**, and the right flange portion **242**. The upper front tab **260** is spaced apart from and faces the upper flange portion **236**. The lower front tab (not shown) is spaced apart from and faces the lower flange portion **238**. In an exemplary embodiment, the first latch member **300** is provided at the left flange portion **240** and the second latch member **302** is provided at the right flange portion **242**. The latch members **300**, **302** may be located proximate to the outer edges of the flange **234** such that the latch members **300**, **302** have a large spacing therebetween.

(26) In an exemplary embodiment, the latch member **300** is a deflectable latch member, which is movable relative to the connector housing **202** during mating to the panel **102**. In an exemplary embodiment, the latch member **302** is a fixed latch member, which is fixed relative to the connector housing **202** during mating to the panel **102**. However, in an alternative embodiment, the latch member **302** may be a deflectable latch member similar to the latch member **300**.

(27) The latch member **300** is formed from the flange **234**. For example, slots **310** are formed in the flange **234** that surround portions of the latch member **300** and allow the latch member **300** to move relative to the flange **234**. The latch member **300** includes a latch arm **312** that extends between a fixed end **314** and a distal end **316**. The latch member **300** includes a latching finger **318** at the distal end **316**. The latching finger **318** extends forward from the latch arm **312**. The latching finger **318** is configured to be received in the panel opening **120** of the panel **102** to latchably secure the latch member **300** to the panel **102**. The latch arm **312** is deflectable relative to the connector housing **202**. For example, the latch arm **312** may be cantilevered from the fixed end **314**. The distal end **316** and the latching finger **318** may be rotated outward during mating with the panel **102**. For example, the latch arm **312** may be hinged at the fixed end **314**.

(28) In an exemplary embodiment, the flange **234** includes an opening **330** defined by inner surfaces **332**. The latch member **300** extends from the flange **234** into the opening **330**. For example, the fixed end **314** defines the connection point for the latch member **300** to the flange **234**. The latch arm **312** occupies the opening **330**. The slots **310** are the portions of the opening **330** surrounding the latch arm **312**. The latch member **300** includes a top **320** and a bottom **322** opposite the top **320**. The slots **310** are located between the inner surfaces **332** and the top **320**, the bottom **322**, and the distal end **316**. In an exemplary embodiment, the inner surface **332** of the flange **234** is located above the top **320** of the latch member **300** to prevent rotation of the latch arm **312** in the opening **330**. For example, the top **320** of the latch member **300** may abut against the inner surface **332** of the flange **234** to prevent rotation of the connector housing **202** relative to the panel **102**. Similarly, the inner surface **332** of the flange **234** is located below the bottom **322** of the latch member **300** to prevent rotation of the latch arm **312** in the opening **330**. For example, the bottom **322** of the latch member **300** may abut against the inner surface **332** of the flange **234** to prevent rotation of the connector housing **202** relative to the panel **102**.

(29) In an exemplary embodiment, the latching finger **318** includes a latching surface **324** configured to engage the inner edge **122** of the panel opening **120** when the electrical connector **200** is in the latched position. The latching finger **318** includes at least one anti-rotation surface different than the latching surface **324**. In an exemplary embodiment, the latching finger **318** includes an upper anti-rotation surface **326** at the top of the latching finger **318** and a lower anti-rotation surface **328** at the bottom of the latching finger **318**. The anti-rotation surfaces **326**, **328** are configured to engage the inner edges **122** of the panel opening **120** to prevent rotation of the connector housing **202** relative to the panel **102**.

(30) In an exemplary embodiment, the flange **234** completely surrounds the latch member **300**. For example, the latch member **300** is embedded in the flange **234**. The flange **234** includes an upper

portion **264** above the latch member **300**, a lower portion **266** below the latch member **300**, and an outer portion **268** outside of the latch member **300**. For example, the outer portion **268** is located between the distal end **316** of the latch member **300** and the first side **220** of the flange **234**. The outer portion **268** connects the upper portion **264** and the lower portion **266** along the outside of the latch member **300**. The structure of the flange **234** provides a solid structure for coupling to the panel **102**. Additionally, the structure of the flange **234**, surrounding the latch member **300** protects the latch member **300** from damage, such as from catching on surfaces or other items, such as the cable during shipping or assembly.

(31) In an exemplary embodiment, the second latch member **302** includes a latching finger **370** (FIG. **3**). The latching finger **370** extends forward of the flange **234**, such as at the right flange portion **242**. The latching finger **370** may be shaped similar to the latching finger **318**. In an exemplary embodiment, the latching finger **370** is fixed relative to the flange **234** (for example, the latching finger **370** is not deflectable relative to the flange **234**). In an exemplary embodiment, the latching finger **370** includes a latching surface **374** configured to engage the inner edge **122** of the panel opening **120** when the electrical connector **200** is in the latched position. The latching finger **370** includes at least one anti-rotation surface different than the latching surface **374**. In an exemplary embodiment, the latching finger **370** includes an upper anti-rotation surface **376** at the top of the latching finger **370** and a lower anti-rotation surface **378** at the bottom of the latching finger **370**. The anti-rotation surfaces **376**, **378** are configured to engage the inner edges **122** of the panel opening **120** to prevent rotation of the connector housing **202** relative to the panel **102**.

(32) FIG. **5** is a sectional view of a portion of the electrical connector **200** in accordance with an exemplary embodiment. FIG. **6** is a top sectional view of a portion of the electrical connector **200** in accordance with an exemplary embodiment. FIGS. **5** and **6** illustrate the contact **204** held in the connector housing **202**. The contact **204** is received in the contact channel **244**.

(33) The contact **204** includes a mating end **270** at a front of the contact **204** and a terminating end **272** at a rear of the contact **204**. The mating end **270** extends into the plug **232** of the connector housing **202**. The mating end **270** is configured to be mated with the mating electrical connector **104** (shown in FIG. **1**). The terminating end **272** is configured to be terminated to the cable **206** (shown in FIG. **2**). In the illustrated embodiment, the terminating end **272** includes a crimp barrel **274** configured to be crimped to the end of the cable **206**. Other types of terminations may be provided at the terminating end **272** in alternative embodiments, such as a weld pad, an insulation displacement contact, and the like. In the illustrated embodiment, the terminating end **272** is located in the base **230** of the connector housing **202**.

(34) The latch member **300** is located outside of the base **230** and the plug **232**. For example, the latch member **300** is provided in the flange **234**. In an exemplary embodiment, a portion of the flange **234** is located outside of the latch member **300** such that the latch member **300** is completely surrounded by the flange **234**. As such, the latch member **300** is protected from damage, such as from catching on surfaces or other items, such as the cable during shipping or assembly.

(35) FIG. **7** illustrates the electrical connector system **100** showing the electrical connector **200** poised for mounting to the panel **102**. FIG. **8** illustrates the electrical connector system **100** showing the electrical connector **200** partially mated to the panel **102**. FIGS. **7** and **8**, and with additional reference to FIG. **1**, illustrate an exemplary mounting procedure for mounting the electrical connector **200** to the panel **102**. In an exemplary embodiment, the electrical connector **200** is a slide-to-latch electrical connector configured to be initially loaded through the panel opening **120** in a loading direction **150** (shown FIG. **7**) to a nominal position (FIG. **8**) and then slid in a latching direction **152** (shown in FIG. **8**) to a latched position (FIG. **1**). The loading direction **150** is generally a forward direction. The latching direction **152** is generally a sideways sliding direction (for example, from right to left in the illustrated view).

(36) During initial loading, the connector housing **202** is rear loaded into the panel opening **120** from behind the panel **102**. The plug **232** of the connector housing **202** is loaded into the main

pocket **130** of the panel opening **120**. The main pocket **130** is oversized relative to the plug **232** to receive the plug **232** and allow the plug **232** to slide sideways in the main pocket **130** from the nominal position (FIG. **8**) to the latched position (FIG. **1**). In an exemplary embodiment, the inner edges **122** defining the main pocket **130** include an upper edge **140**, a lower edge **142**, and side edges **144**, **146**. The side edges **144**, **146** have a width greater than the width of the plug **232** to allow the sideways sliding movement from the nominal position to the latched position. Optionally, the upper edge **140** and the lower edge **142** have a height therebetween approximately equal to a height of the plug **232** to limit up and down movement of the plug **232** within the main pocket **130**. The upper edge **140** and the lower edge **142** guide the sliding movement of the connector housing **202** from the nominal position to the latched position.

(37) In an exemplary embodiment, the tab pockets **136** are provided above and below the main pocket **130**. For example, notches are formed in the upper edge **140** and the lower edge **142** to form the tab pockets **136**. The tab pockets **136** are open to the main pocket **130**. In an exemplary embodiment, the upper and lower tab pockets **136** are offset from each other. Alternatively, the upper and lower tab pockets **136** may be aligned with each other on opposite sides of the main pocket **130**. The tab pockets **136** receive the front tabs **260** of the connector housing **202**. The front tabs **260** are loaded through the panel opening **120** through the tab pockets **136**. In the nominal position (FIG. **8**), the front tabs **260** are aligned with the tab pockets **136** and located forward of the tab pockets **136**. In the latched position (FIG. **1**) the front tabs **260** are offset from the tab pockets **136**.

(38) In an exemplary embodiment, the panel **102** includes a retention tab **148** located immediately adjacent the corresponding tab pocket **136**. The retention tab **148** is positioned to the side of the tab pocket **136** and the direction that the electrical connector **200** slides from the nominal position to the latched position. The front tabs **260** are aligned with, and engage, the retention tabs **148** when the electrical connector **200** is slid sideways from the nominal position. The retention tab **148** is configured to be received in the gap **262** between the front tab **260** and the flange **234**. The retention tab **148** is captured between the front tab **260** and the flange **234**. The retention tab **148** prevents removal of the electrical connector **200** from the panel **102**. For example, the front tab **260** engages the front surface **110** of the panel **102** along the retention tab **148** to prevent removal of the electrical connector **200** from the panel opening **120**. The flange **234** engages the rear surface **112** of the panel **102** to prevent removal of the electrical connector **200** and through the panel opening **120**.

(39) In an exemplary embodiment, the latch pockets **134** are provided at the left side and the right side of the main pocket **130**. For example, notches are formed in the side edges **144**, **146** to form the latch pockets **134**. The latch pockets **134** are open to the main pocket **130**. In the illustrated embodiment, the latch pockets **134** include a first latch pocket **134***a* and a second latch pocket **134***b*. The first latch pocket **134***a* receives the first latch member **300**. The second latch pocket **134***b* receives the second latch member **302**. In an exemplary embodiment, the second latch member **302** is received in the second latch pocket **134***b* in the nominal position (FIG. **8**) and slides sideways in the second latch pocket **134***b* to the latched position (FIG. **1**). In the illustrated embodiment, the second latch member **302** is a fixed latch member; however, the second latch member **302** may be a deflectable latch member in alternative embodiments. In such embodiment, the second latch member **302** may be latched into a different panel opening in the panel **102**. The second latch pocket **134***b* includes an upper edge **160**, a lower edge **162**, and a side edge **164**. The upper anti-rotation surface **376** of the latching finger **370** faces the upper edge **160** and may engage the upper edge **160** to prevent rotation of the electrical connector **200** relative to the panel **102**. The lower anti-rotation surface **378** of the latching finger **370** faces the lower edge **162** and may engage the lower edge **162** to prevents rotation of the electrical connector **200** relative to the panel **102**. The upper and lower edges **160**, **162** may guide the latching finger **370** as the electrical connector **200** is slid sideways from the nominal position to the latched position.

(40) In an exemplary embodiment, in the nominal position (FIG. **8**) the latching finger **318** of the first latch member **300** is located behind the rear surface **112** of the panel **102**. The latch member **300** is deflected rearward as the flange **234** of the connector housing **202** presses against the rear surface **112** of the panel **102**. For example, because the latching finger **318** extends forwardly, when the outer surface of the latching finger **318** engages the rear surface **112** of the panel **102**, the latch arm **312** is deflected rearwardly. The electrical connector **200** is slid sideways from the nominal position to the latched position (FIG. **1**). In the latched position, the latching finger **318** is aligned with the latch pocket **134***a*. The latch member **300** snaps forward to position the latching finger **318** in the latch pocket **134***a*. The latch finger **318** engages the inner edge **122** defining the latch pocket **134***a* to latchably couple the electrical connector **200** to the panel **102**. The latching surface **324** engages the inner edge **122** to prevent sliding of the electrical connector **200** in the opposite direction (for example, toward the nominal position).
(41) In an exemplary embodiment, the first latch pocket **134***a* includes an upper edge **170**, a lower edge **172**, and a side edge **174**. The latching surface **324** engages the side edge **174** to prevent sliding of the electrical connector **200** relative to the panel **102**. The upper anti-rotation surface **326** of the latching finger **318** faces the upper edge **170** and may engage the upper edge **170** to prevent rotation of the electrical connector **200** relative to the panel **102**. The lower anti-rotation surface **328** of the latching finger **318** faces the lower edge **172** and may engage the lower edge **172** to prevent rotation of the electrical connector **200** relative to the panel **102**.
(42) The electrical connector **200** is latchably coupled to the panel **102** in the latched position. The electrical connector **200** is mounted to the panel **102** without the use of tools or mounting hardware. In an exemplary embodiment, the electrical connector **200** is a slide-to-latch electrical connector configured to be loaded through the panel opening **120** to a nominal position and then slid in a latching direction to a latched position. Integral features of the electrical connector **200**, such as the latch members **300**, engage the panel **102** to secure the electrical connector **200** to the panel **102**.
(43) It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from its scope. Dimensions, types of materials, orientations of the various components, and the number and positions of the various components described herein are intended to define parameters of certain embodiments, and are by no means limiting and are merely exemplary embodiments. Many other embodiments and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112 (f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

Claims

1. An electrical connector for mounting to a panel, the electrical connector comprising: a connector housing having a base at a rear of the connector housing, a flange extending from the base, and a plug extending forward of the base and the flange at a front of the connector housing for mating with a mating electrical connector, the connector housing having a contact channel through the base

and the plug; a contact received in the contact channel, the contact having a mating end and a cable end, the cable end of the contact configured to be terminated to a cable; a latch member integral with the connector housing and located within the flange for latchably coupling to the panel, the latch member including a latch arm located within the flange being deflectable relative to the connector housing, the latch member including a latching finger extending forward from the latch arm to be received in a latch pocket of the panel to latchably couple the latch member with the panel; wherein the plug of the connector housing is configured to be inserted into a panel opening of the panel in a loading direction, the connector housing being movable laterally relative to the panel in a latching direction to a latched position, the latching direction differing from the loading direction, the plug movable in the panel opening in the latching direction, the latching finger engages an inner edge of the panel opening when the housing is in the latched position to hold the connector housing in the latched position.
2. The electrical connector of claim 1, wherein the flange surrounds the latch member.
3. The electrical connector of claim 1, wherein the flange includes an opening, the latch arm located in the opening and movable relative to the flange in the opening.
4. The electrical connector of claim 1, wherein the latch member includes a top, a bottom, a fixed end, and a distal end opposite the fixed end, the fixed end positioned closer to the base than the distal end, the fixed end extending from the flange, the flange surrounding the top, the bottom, and the distal end.
5. The electrical connector of claim 1, wherein the latching finger extends forward of the flange for reception in the panel opening of the panel.
6. The electrical connector of claim 1, wherein the latching finger includes a latching surface configured to engage the inner edge of the panel opening in the latched position, the latching finger including at least one anti-rotation surface different than the latching surface configured to engage the inner edge of the panel opening in the latched position, the at least one anti-rotation surface preventing rotation of the connector housing relative to the panel.
7. The electrical connector of claim 1, wherein the flange is provided at a first side and a second side of the connector housing, the latch member provided at the first side, the electrical connector further comprising a second latch member at the second side.
8. The electrical connector of claim 7, wherein the second latch member includes a second latch arm being deflectable relative to the connector housing.
9. The electrical connector of claim 1, wherein the connector housing includes a front tab configured to pass through the panel opening to engage a front surface of the panel, the flange configured to engage a rear surface of the panel, the connector housing being configured to capture the panel between the front tab and the flange.
10. The electrical connector of claim 9, wherein the front tab is an upper front tab at a top of the connector housing, the connector housing further comprising a lower front tab at a bottom of the connector housing.
11. The electrical connector of claim 1, wherein the connector housing is manufactured from a dielectric material, the latch member being manufactured from the same dielectric material as the connector housing.
12. The electrical connector of claim 11, wherein the connector housing and the latch member are co-molded using the dielectric material.
13. The electrical connector of claim 1, wherein the plug includes a slot configured to receive a bus bar.
14. An electrical connector system comprising: a panel having an inner edge defining a panel opening, the panel opening including a main pocket and a latch pocket extending from the main pocket; and an electrical connector mounted to the panel, the electrical connector including a connector housing having a base at a rear of the connector housing, a flange extending from the base, and a plug extending forward of the base and the flange at a front of the connector housing

for mating with a mating electrical connector, the connector housing having a contact channel through the base and the plug, the electrical connector including a contact received in the contact channel having a mating end and a cable end configured to be terminated to a cable, the electrical connector including a latch member integral with the connector housing and located within the flange for latchably coupling to the panel, the latch member including a latch arm located within the flange being deflectable relative to the connector housing, the latch member including a latching finger extending forward from the latch arm to be received in a latch pocket of the panel to latchably couple the latch member with the panel; wherein the plug of the connector housing is inserted into the main pocket of the panel opening in a loading direction, the connector housing being movable laterally relative to the panel in a latching direction to a latched position, the latching direction differing from the loading direction, the plug movable in the main pocket in the latching direction, the latching finger being received in the latch pocket in the latched position and engaging the inner edge of the panel opening in the latched position to hold the connector housing in the latched position.

15. The electrical connector system of claim 14, wherein the latching finger slides along a rear surface of the panel as the connector housing is moved from the loaded position to the latched position, the latching finger moving forwardly into the latch pocket when in the latched position.

16. The electrical connector system of claim 14, wherein the latch pocket is at a first side of the main pocket, the panel opening including a second latch pocket at a second side of the main pocket, the electrical connector including a second latch member received in the second latch pocket.

17. The electrical connector system of claim 14, wherein the flange surrounds the latch member, the flange engaging a rear surface of the panel around the inner edge defining the latch pocket.

18. The electrical connector system of claim 14, wherein the flange includes an opening, the latch arm located in the opening and movable relative to the flange in the opening.

19. The electrical connector system of claim 14, wherein the latch member includes a top, a bottom, a fixed end, and a distal end opposite the fixed end, the fixed end positioned closer to the base than the distal end, the fixed end extending from the flange, the flange surrounding the top, the bottom, and the distal end.

20. The electrical connector system of claim 14, wherein the latching finger extends forward of the flange for reception in the latch pocket of the panel opening.